

CO3-AT-2

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age

0
in Come

student

credit rating

buy -
Computes

2=30

high

no

fair

No

<=30

high

no

31.400

high

740

740

CO10

31... 90

4=30

740

medium

Low

Medium

high

no

2255 25

yes

excellent

NG

fair

yes

yes

excelent

yes

fail

yes

fail

No

excellent

No

T

yes

$$\sum_{i=1}^n D_i \log (p_i)$$

9=1

Entropy:

$$H(Y) =$$

$$\text{Info } (D) =$$

Info A (0)

=

$$-\sum_{P=0}^m PP \log_2 (PP)$$

1081 J = 1 1D1

X info (DS)

$$\text{Gain (A)} = \text{Info (D)} - \text{InfoA (D)}$$

Assume

Class P is

who is

buy-computer =
"yes"

Class N

is

not buy computer =

Table:

88

class

Age

Yes

no

P

N

$I(P, N)$

2=30

2

co

0.971

31-40

I

a

O

>HO

3

2

9

5

d b

O

0.971

0.940

$$I(2,3) = -\{pp \log(P)$$

I (4, 0)

-2

3 1092

■ 끝 10g (증) - 음 182
(춤)

=

5

-2 (-1.322) - 32
(-0.737)

음

$$0.5288 + 0.4422$$

$$= 0.971$$

$$- \text{Epilog (PP)}$$

$$-4$$

$$= 4$$

$$1092 (1) - 04$$

$$1092 (0/4)$$

$$I(3, 2) = - \sum \text{pilog} \\ (\text{PP})$$

$$3$$

$$\log$$

$$= -2 / = (\log_2 (232))$$

$$- \frac{2}{2} 10g^2 (232)$$

M

$$0.4422 + 0.5288$$

$$(0-971$$

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$$I(9,5)$$

5

11

$$\begin{aligned} & \text{유 } 10g, (\text{유}) - \text{유} \\ & 1032 (\text{휴}) \end{aligned}$$

14

-9

14

14

$$(-0.6374) -$$

5

$$- \frac{1}{14} (-1.4854)$$

$$0.4097 + 0.5304$$

$$= 0.940$$

info

$$\text{age (D)} = 5$$

$$\times 0.971 +$$

$$\frac{1}{14}$$

$$14$$

$$14$$

$$\frac{40 + 5 \times 0.971}{14}$$

$$0.694$$

information gain

(age)

$$= 0.940 -$$

$$0.694$$

$$0.246$$

for the attribute income.

Table:-

Income	Yes	NO	$I(P, N)$
	P	N	
High	2	2	0.971
Medium	4	2	
Low	3	5	0.811
			0.940

$$I(2,2) = \text{Epilog}(pt)$$

11

$$- \frac{2}{4} 1092 (24) - \frac{2}{4} 1092 (2) \\ (\text{증})\text{-류}$$

$$== \text{증} (-1)$$

$$-\text{증}(-1)$$

$$\text{류} (-1)\text{-쥬너})$$

11

=

$$22 = 13$$

$$I (4,2) = -E_{pp} \log \\ (PP)$$

$$=-4 \log a (4) - 2 \log a \\ (2)$$

중1

0

$$-\frac{\text{뵚}}{\text{중}}(-0.585) - \frac{\text{중}}{\text{뵚}}(-1.585)$$

$$0.389 + 0.528$$

$$0.917$$

$$l(3,1) = - \operatorname{Epilog}(\pi)$$

ॐ

$$-2/109(4) - 1092(+)$$

$$0.311 + 0.5$$

$$0.811$$

Info

income

$$(D) = \{ 1DP1$$

$$\times Pn fo (D)$$

$$4 \times 0.811$$

14

IDI

+

$$xo-917 xyXI$$

TI

14

$$=0.231 +0.285 +0.392$$

098)

Grain Income

=

$$0.940 -0.9111$$

0029

Gain (student)

$$=0.151$$

$$\text{Gain (credit rating)}=0.048$$

$$\text{gain (A)/Splitinfo (A) ,}$$

$$\text{Grain ratio} =$$

$$\text{Split Info}$$

$$\text{A (D) =}$$

$$\text{V1D91}$$

$$\text{x log 2}$$

$$\text{3=1 1P1}$$

$$\text{Split infoage} = -2 \text{ (as)}$$

--

$$\text{도 } 10g2 \text{ 트}$$

$$14$$

$$101$$

$$\text{-5 109. 5-4 109.}$$

$$(4)$$

$$14$$

$$14$$

$$5$$

14

1092
(54)

= -5

(-1-4854) - 4

(-1.8074)-5

14 (-1-4

854)

14

14

(-1-4854)

= 1.58.

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Gain Ratio

= Gown

Split

Split

income

= 0.246

1.58.

ID 1

$$\begin{aligned} & \quad \quad \quad -4 \\ & \quad \quad \quad 14 \\ = & -14 \ 1093 \ (14) \ + \ 6 \\ & 1093 \ (\ 14 \) - 14 \ 103 \\ & (47) \end{aligned}$$

11

14

14

$$\begin{aligned} & - \ 4 \ (-1.8074) - 6 \\ & (-1-8220) - | - | - | \end{aligned}$$

14

11
=

GR

= Grain

Split

info

1-557

(-1-8074)

فرمة

م مدة

0.019 Whisnoj Query

Gini Index

Gini (D) = 1 - $\sum_{i=1}^n p_i^2$

finding gini indexe
Index based on income

Gini (D) = 1 - $\sum_{i=1}^n p_i^2$

Pg 2

20

-4

10

M

- 6

H

-U

4-4

14

Стай

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(D) = 1 -

•

[(452

+

(12)"]

[13

C

O

$$0.413 + 0.127$$

127
]

$$1 - 0.54$$

$$= 0.46$$

Partition of data set

with respective income.

(low, medium) -
10

D_i

D₂ =

(High)

4

gini income (D)

=

11

10)

gini (D,) + 1

D₂)_gini (D₂)

(D)

= 10

14 [1 [0 . 4 9] +

[0.09]]) + 44 [1-[0.25

+0.25]

$$\star \frac{10}{14}$$

$$[1 - 0.58.3 +$$

$$4/4 [1-0.50]$$

$$14$$

$$10 [0.42] + 4$$

$$[0.50]$$

$$14$$

$$= 0.299 + 0.142$$

$$0-443$$

$$14$$

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2

Actual class / Predicted class

cancel =yes

Cancer NO

Total

Cancel = yes	90	210	300
cancel = NO	140	9560	9700
товал	230	9770	10000

$$1 \text{ Accuracy} = (TP + TN) / \text{All}$$

$$\frac{90+9560}{10000} = 0.965$$

ii) Sensitivity

$$= \frac{TP}{P}$$

$$= 90/230$$

$$= 0.3913$$

Pii, Error

rate

1

Accuracy

1-0.965

= 0.035

iv) Specificity =

TN/N

9560/9

770

= 0.985

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v

precis

ion

=

TP

V_{ij}
recall

V_i
 $V_9) F_i$

$TP + FP$

$$= \frac{90}{90 + 140}$$

TP

Why $PP + FN$

$$= \frac{90}{90 + 140} = 0.391$$

$$\begin{aligned}
 & \frac{.90}{90 + 210} = 0.3 \\
 & = 2 \times \frac{0.391}{0.391 + 0.3} \times 0.3 \\
 & = 0.3395 \\
 & \quad 0.2346 \\
 & \quad 0.691
 \end{aligned}$$

0A Spheaphored bog
get

get bored tending the town's flock to have
some fun he cries out, "wolf" even though no welf

in sign The villages dun the protect the flock
lagt then get rellaly mad when the
mad when the realize the boy was
playing on them one night, the shepherd
boys see

a доке

a

real wolf approaching the flow and call out "
wolf" The villages refuse to be fooled
again and stay in home

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Act

W=X

Pred

Wolf = Y

N

тп

FN

FP.

TNI

2. Assume there

are

100 images, 30 of them depict a cat, A machine

learning model predicts the

rest do not. A machine

Occurrence

absence

cat

of

of cat

the

in 25 of 30 cat images. It also predict Pn 50 of the no cat Pimages.

to

Identify TP, FP, FN, TN and calculate the
precision and Recall

and F1 score

cat = yes

cat =NO

Cat=yes

Cat No

TP: 25

=

Correctly Predict

Fp=20

No cat exists

FN=5

Cat exists

TN=50

Correctly
Predicted

1.)

Precision

Precision

=

tp

11

25

25+20

TPTFP

= 25

45

=0.556

Sum

Recall

Recall =

TP

TP+FN

25

25+5

= 25

30

= 0.8333

3 FI score

$$f_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$2(0.556) - (0.8333)$$

$$0.556 + 0.8333$$

$$0.667$$

as

Service

Provider want to
clausity

Spam or not spam using features.

An

ema

email

9 P (spam)

$$3 = 22$$

글

P (not spam)

$$3 = =$$

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is conditional
probabilities

$$P(0 = 4/spam) = 2/3$$

$$P(0 = No / spam) = \frac{1}{3}$$

$$P(free = 4/spam) = 3/3$$

$$P(Free = No / Spam) = 0/3$$

$$P(0=Y/Not$$

Spam) = 1/3
P(0
=/NO/Notspa
m)=2/3

4

)
999, Naive Bayes.

Feature

condition

Offer

Spam

yeshiva 2/3

not spam

Offer

NOVY

y₃

213

Free

yes

313=1

01320

Free

No

01320

3/3=1

Length

Short

213

1/3

Length

Song

43

2/3

Spam =

x

32x2x0x35

0

6

=0

Since

0.0556 > 0

(offer = yes,

Free - No

Moid

|

7/4/2

0

not/spam = = = =

= = = = =

1/

+18=0.05

56

Length = Short) - not
spam